Formula 1 Podium Prediction

Predicting top-three finishes from pre-race information

Analytical report

CSEN 903 · Advanced Computer Lab
Formula 1 World Championship dataset, 1950–2024

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1. Summary

For every driver in every race, we predict whether the driver finishes on the podium (1st, 2nd or 3rd), using only information known after qualifying and before the race starts. The raw relational tables were audited and cleaned into a table with exactly one row per driver per race, three data-engineering questions were answered, leakage-safe features were built, and three models were compared against a grid-only baseline.

- **Data:** 26,668 driver–race entries (1950–2024) after resolving 91 repeated records from shared drives in 1950–1978.
- **Features (10):** starting grid, qualifying position and Q1 gap; driver and team past podium rates and championship points share before the race; and three circuit features — all computed from information available before the start.
- **Split by season:** training 1980–2019 (start year chosen on validation), validation 2020–2021, test 2022–2024 (used once, at the end).
- **Results on the unseen test seasons (PR-AUC, the primary metric):** grid-only baseline 0.647, logistic regression 0.676, gradient boosting 0.715, shallow neural network 0.698. All three models beat the baseline. The neural network was best on validation (0.756); gradient boosting was best on test.
- **What drives the predictions:** starting grid position is by far the most important feature for every model, followed by driver and team form. Qualifying and team points add little once grid and driver form are known. These are associations the models use, not causes of podium finishes.

2. Problem and data

Unit of analysis: one driver entering one race. **Target:** *podium* = 1 when the final classified position (*positionOrder*) is 1–3, otherwise 0. *positionOrder* is used because it is filled for every entry, including drivers who retired. **Prediction cutoff:** after qualifying and before the race starts. The current race's results, status, points, laps, pit stops and post-race standings are never used as inputs.

The Kaggle/Ergast dataset contains 14 linked tables. All 14 received structural checks (unique non-empty IDs and links that point to existing records). The nine tables needed for the analyses and models received detailed exploration and cleaning (Table 1).

Table	Rows	Detailed cleaning	Table	Rows	Detailed cleaning
races	1,125	Yes	driver_standings	34,863	Yes
results	26,759	Yes	seasons	75	No
circuits	77	Yes	constructor_results	12,625	No
drivers	861	Yes	lap_times	589,081	No
constructors	212	Yes	pit_stops	11,371	No
status	139	Yes	sprint_results	360	No
qualifying	10,494	Yes	constructor_standings	13,391	Yes

Table 1. *The 14 source tables. Tables marked "No" are kept as loaded and are not model inputs.*

Podiums are rare: about 12.7% of all entries (3 of roughly 20–30 cars per race) and about 15% in recent seasons, so the task is an imbalanced binary classification problem.

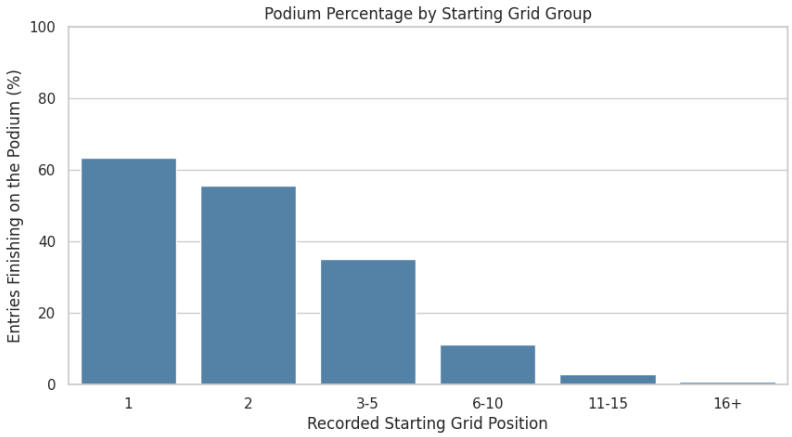


Figure 1. *Original data: share of entries finishing on the podium by starting grid group. The grid position is the strongest single signal.*

3. Data cleaning and audit

3.1 Cleaning rules

- **Missing-value markers:** the literal text *N* (147,395 cells in the nine core tables) was converted to real missing values. Nothing was filled in at this stage.
- **Semantic typing:** dates parsed as dates; lap times (M:SS.sss) converted to seconds; numeric result fields converted to numbers; car numbers stored as text (labels, not quantities). Unexpected formats stop the code instead of disappearing silently.
- **Labels:** leading/trailing spaces removed from nationality and country labels; historical team names and IDs are not merged.
- **Grid 0:** kept as recorded. The audit showed it mostly marks entries that did not qualify, did not pre-qualify or withdrew, plus pit-lane starts; for modeling it is treated as "no numbered grid slot" (Section 7).
- **No deletions to tidy charts:** missing Q2/Q3 times (drivers knocked out earlier) and very long Q1 times are kept and reported rather than removed.

3.2 One row per driver per race

85 driver–race combinations appeared more than once (176 records, 91 extra rows, 0.34% of results). All of them are from 1950–1978, 76 of them from 1950–1957, when drivers shared or swapped cars during a race. In 21 groups one record was a podium and another was not. **Rule:** keep the record with the best finishing order (ties: lowest *resultId*), so a driver who reached the podium in any of their cars keeps that podium. Because the kept row can come from a different car than the one the driver started in, these seasons are a limitation for historical pre-race features; none of them falls inside the final training window (1980–2019).

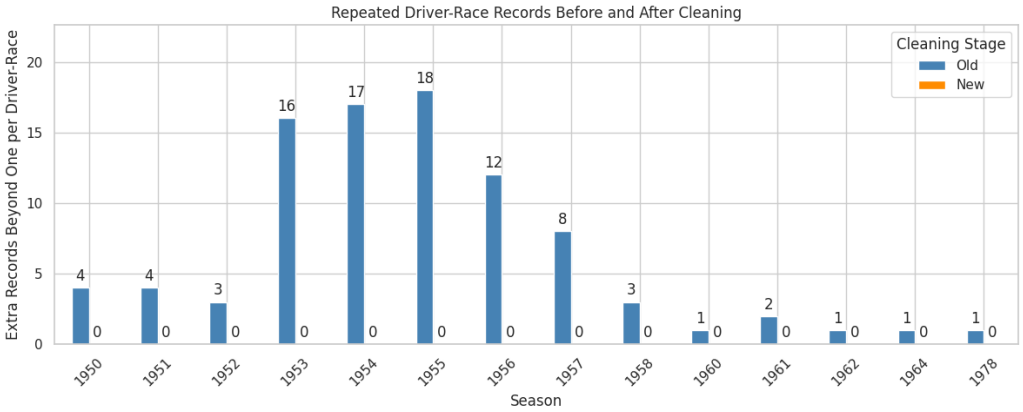


Figure 2. *Extra records beyond one per driver–race, by season, before (blue) and after (orange) cleaning.*

3.3 Effect of cleaning

Measure	Before	After	Meaning
Result rows	26,759	26,668	91 extra rows removed; every driver–race kept once
Extra rows for an existing driver–race	91	0	Grain fixed to (raceId, driverId)
\N markers in the nine core tables	147,395	0	Converted to real blanks, not filled
Rows labelled as a podium	3,397	3,395	A podium stored twice is counted once
Share of rows that are podiums	12.695%	12.731%	+0.036 percentage points
Largest change in podium rate of a grid group	—	0.678 pp	Practically unchanged
Largest change in qualifying coverage of a season	—	0.000 pp	Unchanged
Value checks with a flagged row	—	0 of 17	No impossible values found

Table 2. *Cleaning summary, calculated in the notebook.*

All primary-key and foreign-key checks passed for all 14 tables, and all 17 validity checks (negative counts, non-positive lap times, races before a driver's birth, etc.) found no flagged rows. The longest recorded Q1 time (about 1,003 s against a 99th percentile of about 124 s) is kept and reported as a limitation.

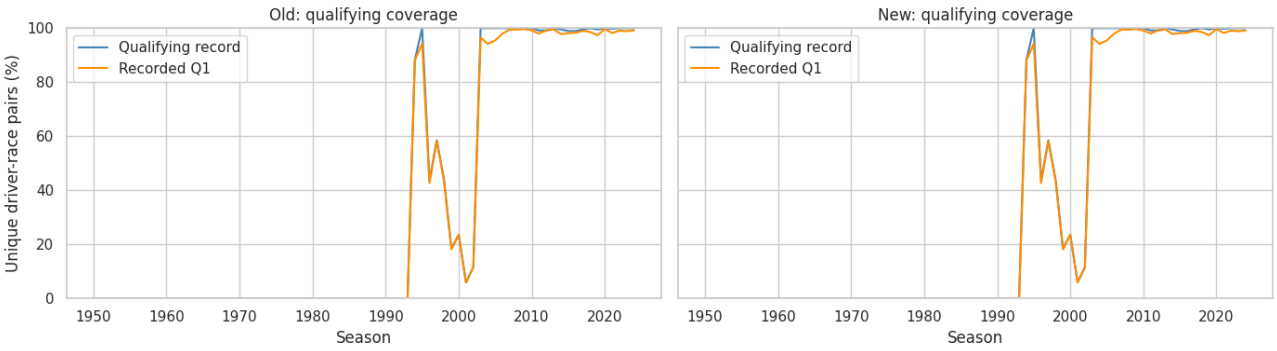


Figure 3. *Share of driver–race pairs with a qualifying record and a Q1 time, by season, before and after cleaning. Qualifying data starts in 1994, is patchy in 1996–2002 and is complete from 2003.*

4. Data-engineering questions

Each question joins only the tables it needs; every join was checked to keep one row per driver per race. Race outcomes are used here to *measure* results, never as predictors.

Q1. Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

Tables: results → races → circuits, plus qualifying. **Grain:** one driver–race; rates per circuit and era. A circuit needs at least 10 front-row entries in an era to be ranked (38 circuit–era groups excluded).

- **Before 2014** (actual grid P1–P2): Magny-Cours 80.6% (n=36), Istanbul Park 78.6% (n=14), Dijon-Prenois 75.0% (n=12).
- **2014 onward:** Suzuka 88.9% (n=18), Yas Marina 86.4% (n=22), Monaco 85.0% (n=20).
- **The ranking changes:** among circuits ranked in both eras, Monaco rose from 15th to 3rd and the Hungaroring fell from 2nd to 16th.
- **Qualifying vs grid:** of 10,382 entries with both positions, 23.3% started from a different position than they qualified (18.2% further forward, 5.1% further back), so the two were analysed separately.
- **Missingness:** 16,174 entries without a qualifying position are left out of the qualifying-based ranking only; grid 0 is left out of grid-based rankings.

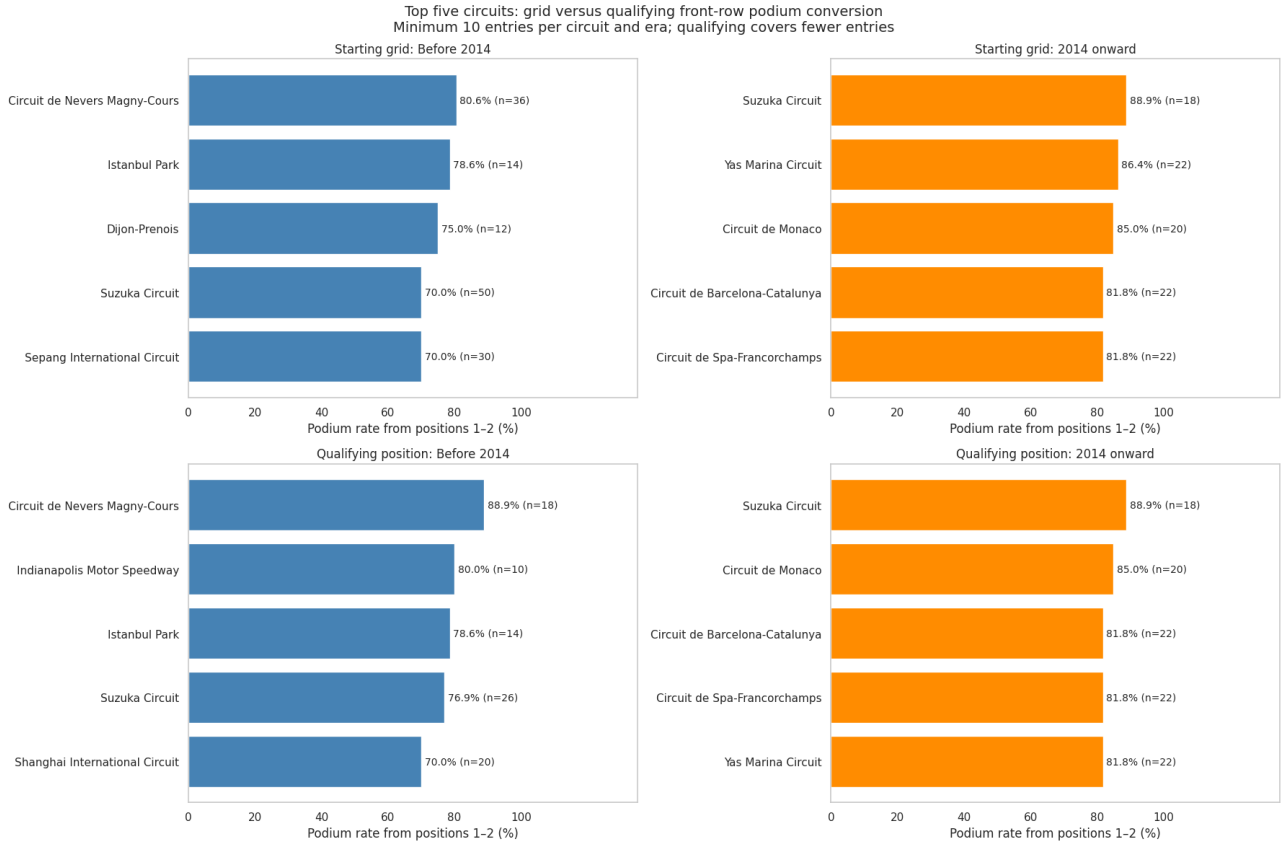


Figure 4. *Top five circuits by front-row podium conversion, using the starting grid (top) and the qualifying position (bottom), before 2014 (left) and from 2014 (right).*

Q2. Do drivers reach the podium more often at their home race, at the same starting position?

Tables: results → races → circuits (country) and drivers (nationality), with an explicit nationality-to-country mapping. **Control:** drivers are compared at the same grid position, and the home and away rates are re-weighted to the same grid mix.

- **No overall home advantage:** adjusted podium rate 13.57% at home vs 13.69% away (−0.12 percentage points).

- **Mixed by grid slot:** from P3, home 54.79% vs away 45.82%; from P6, home 13.75% vs away 19.20%. Home groups are small (about 70–90 entries per slot), so individual slots are noisy.
- **Exclusions:** 381 Indianapolis 500 entries (a largely separate driver population), 4 entries with dual nationalities (no unambiguous country), 1,635 entries with zero or negative grid, and 79 explicit non-starts.
- Matching on grid does not control for team, driver or year, so the result does not prove that racing at home has no effect.

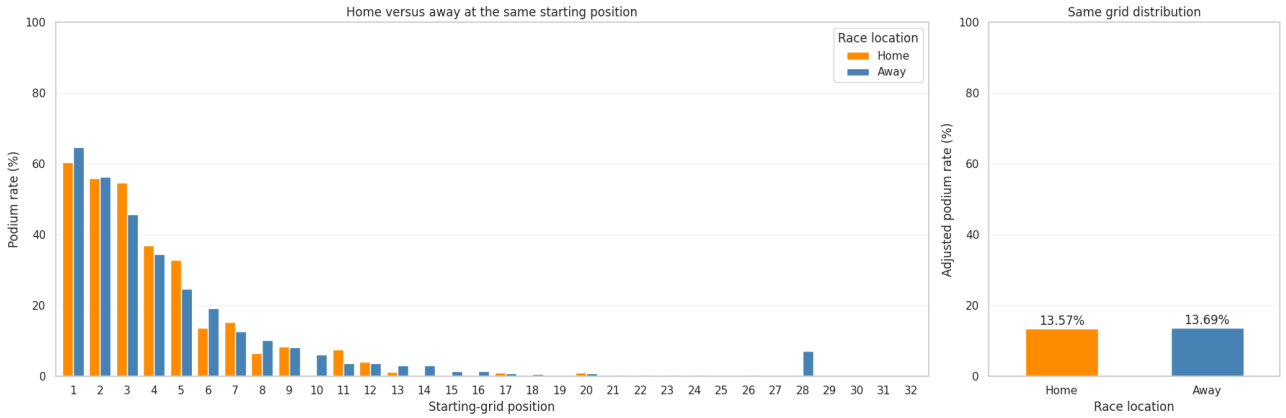


Figure 5. *Podium rate at each grid position for home and away drivers (left) and the grid-adjusted overall rates (right).*

Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

Tables: results → status → constructors → races. A **mechanical retirement** is a start that ended because a car component failed (engine, gearbox, brakes, power unit, hydraulics, etc.). Accidents, collisions, driver errors and regulation decisions are not counted, and vague labels ("Retired", "Technical") are excluded. Explicit non-starts are removed from the denominator (7 and 4 entries); the 97 and 32 zero-lap entries are kept because they include first-lap failures, so starts are estimates.

- **2014–2021:** Caterham 23.5% (8 failures in 34 starts), Lotus F1 17.3% (n=75), Toro Rosso 16.1% (n=242).
- **2022–2024:** Alfa Romeo 11.4% (n=88), Alpine 11.0% (n=136), AlphaTauri 6.8% (n=88).
- Rates fell broadly in the newer era. Caterham's figure rests on few starts and is the least certain. Engine failures are the most common mechanical status in both eras.

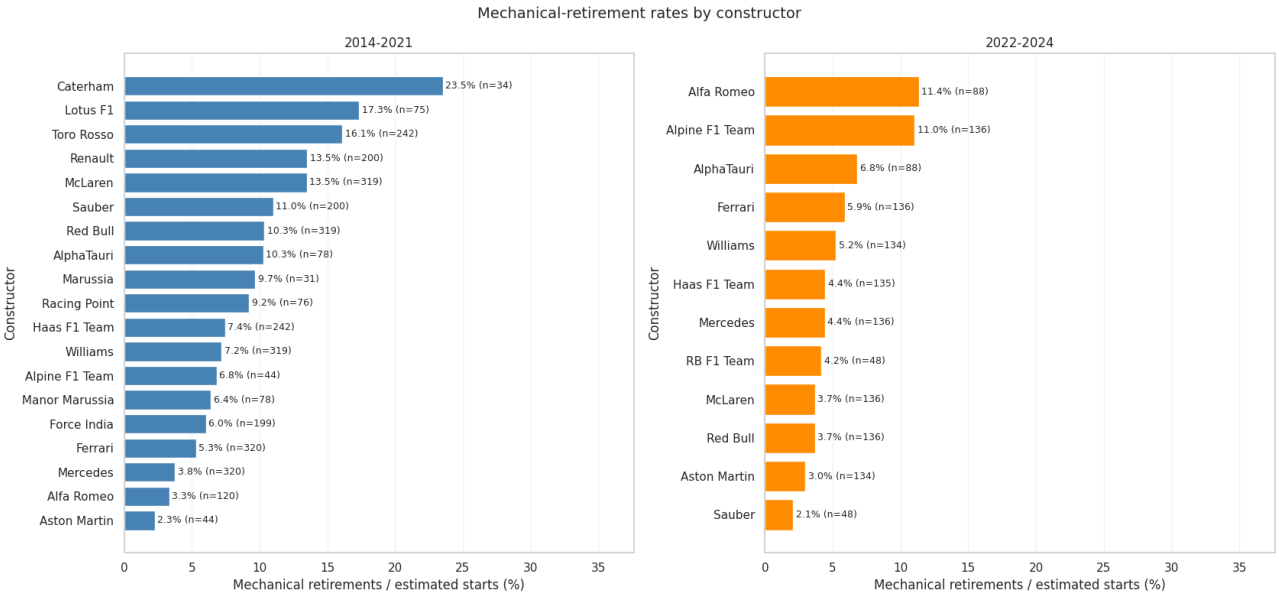


Figure 6. *Mechanical-retirement rate by constructor and era, with the number of estimated starts.*

5. Information available at the pre-race cutoff

Every source field was judged against the cutoff (after qualifying, before the start). Fields that are only known once the race has started are never model inputs. Fields that are stored after the race but describe the past (championship standings) are made safe by looking only at the previous round.

Source	Judgement	How it is handled
results	Mixed	Grid is safe (fixed before the start). Finishing position, points, laps, time, fastest lap and status are race outcomes and are never inputs; earlier races' results are used only through past-only rates.
races, circuits, drivers, constructors	Safe	Calendar and fixed facts; used for ordering, splitting and building past-only features.
qualifying	Safe	Takes place before the race; qualifying position and Q1 gap are used. Q2/Q3 are not, because most drivers have no Q2/Q3 time.
driver_ / constructor_standings	Leaky as stored	Totals after each race include that race. Made safe by using the standings after the previous round of the same season.
status, lap_times, pit_stops, constructor_results	Leaky	Describe the race itself; used only for the retrospective analysis (Q3), never as inputs.
sprint_results	Judgement call	Technically pre-race, but only 18 weekends exist; in 2021–2022 the sprint set the Sunday grid, so its effect is already in the grid; excluded for a consistent feature set across seasons.

Table 3. *Leakage judgement for each source.*

6. Selected features and why each is valid

Raw driver, team and circuit IDs were not used directly: most IDs are rare (749 of 861 drivers have fewer than 80 entries), and several recent circuits never appear in the training seasons. Each was replaced by a number calculated from **earlier races only**. Races are processed in date order; every entry on a date is encoded from earlier dates, and that date's results are added to the history afterwards, so neither the driver's own result nor a team-mate's result can leak in.

Feature	Definition	Why it is valid before the race
grid	Starting position after penalties (0 = pit lane → missing)	Fixed before the start
qualifying_position	Qualifying result	Qualifying happens before the race
q1_gap_percentage	% slower than the fastest Q1 time in that race (comparable across circuits)	Qualifying happens before the race
driver_podium_rate_pre_race	Driver's podium rate in all earlier races, smoothed towards the overall rate (10 pseudo-races)	Uses earlier dates only
constructor_podium_rate_pre_race	Same for the team	Uses earlier dates only
driver_points_share_pre_race	Driver's points after the previous round ÷ leader's points (comparable across scoring systems)	Previous round only; 0 at season openers
constructor_points_share_pre_race	Same for the team (missing before 1958)	Previous round only
circuit_front_row_conversion_pre_race	Share of earlier P1–P2 starters at this circuit who reached the podium (links to Q1)	Earlier races at the circuit only
driver_circuit_lift_pre_race	Driver's earlier podium rate here minus their own rate (0 on a first visit)	Earlier races only
constructor_circuit_lift_pre_race	Same for the team	Earlier races only

Table 4. *The ten selected features (Trial 3).*

Evidence and nested trials. The features were organised into three nested feature sets, each tested later on validation: **Trial 1** grid + qualifying (3 features); **Trial 2** + driver and team form (7); **Trial 3** + circuit features (10). Plots of the podium rate per feature range (2014–2024) supported each group before training:

- **Grid and qualifying:** about 80% podium rate from P1, 71% from P2, 55% from P3, under 5% from P7 onward. Spearman correlation with podium: qualifying −0.52, grid −0.49, Q1 gap −0.41.
- **Form:** the championship leader reaches the podium about 78% of the time; driver points share correlates 0.46 with podium (raw points about 0.40). At the same grid group, cars from stronger teams reach the podium far more often (from P4–6: 28% vs 15% vs 7%).
- **Circuit:** weak evidence. Front-row conversion has no clear pattern and the lifts correlate only 0.01–0.07 with podium; they were kept as a fair test and judged on validation.

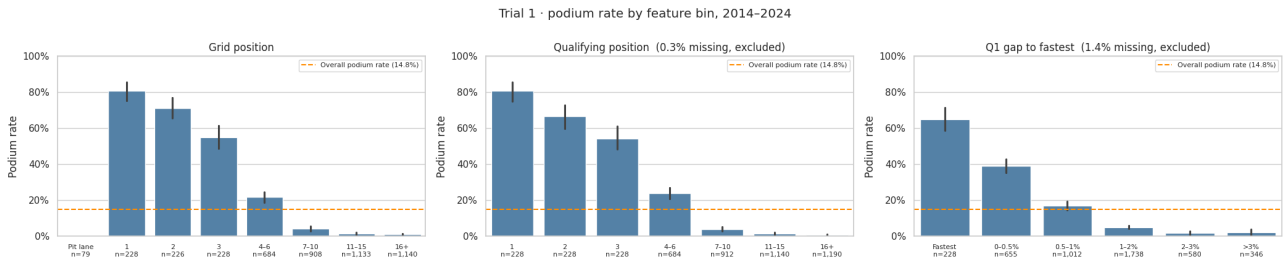


Figure 7. *Trial 1: podium rate by grid position, qualifying position and Q1 gap (2014–2024), with 95% confidence intervals and group sizes.*

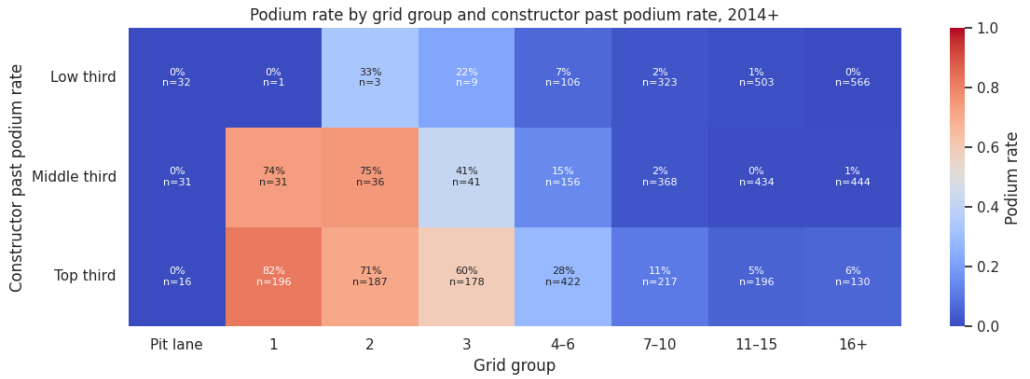


Figure 8. *Trial 2: podium rate by grid group and team strength (thirds of past podium rate). Form adds information that the grid does not contain.*

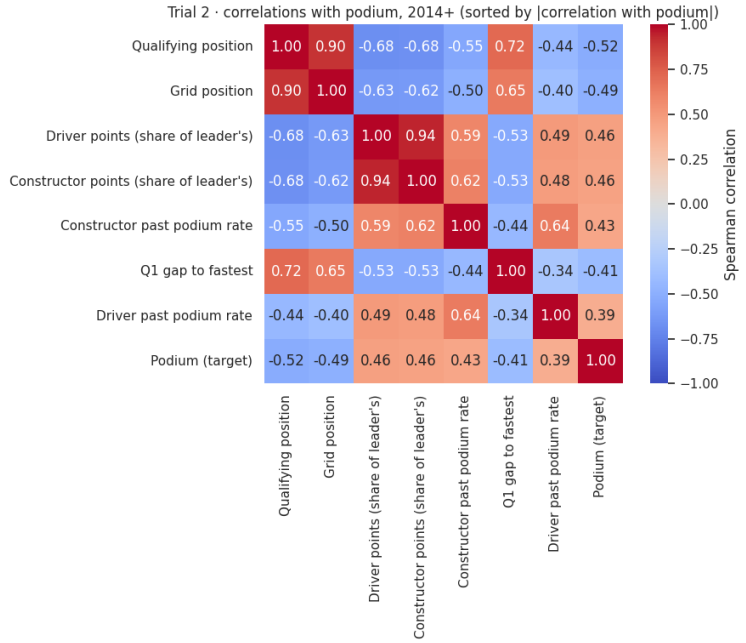


Figure 9. *Trial 2: Spearman correlations of the features with each other and with podium (2014–2024).*

7. Splitting and preprocessing

7.1 Season-based split

The data is split by time because the model will always predict races after the data it learned from. A random split would let it learn from 2023 to "predict" 2015.

Set	Seasons	Entries	Podiums	Podium rate	Use
Training	1980–2019*	16,693	2,070	0.124	Fit models and preprocessing
Validation	2020–2021	780	117	0.150	All choices: start year, features, order, settings, thresholds
Test	2022–2024	1,359	204	0.150	Used once for the final scores

Table 5. *Split used by the final models. *The training start year was chosen on validation (Section 7.3); the candidate pool covers 1950–2019 (24,529 entries).*

7.2 Preprocessing and its effect

Two steps are applied: **median imputation** (a missing value is replaced by the training median) and **standard scaling** (each feature shifted and stretched to mean 0 and standard deviation 1). Both are fitted on the training rows only and then applied unchanged to validation and test. Grid 0 (pit-lane start) is treated as missing so it is not read as "better than pole". After the steps, no values are missing and every feature has mean 0 and spread 1. Q1 gap stands out: 66.5% of its training values were missing (seasons before the mid-1990s) and became the median, and a few extreme gaps (up to 923%) inflate its spread (10.35), squeezing normal values into a narrow band.

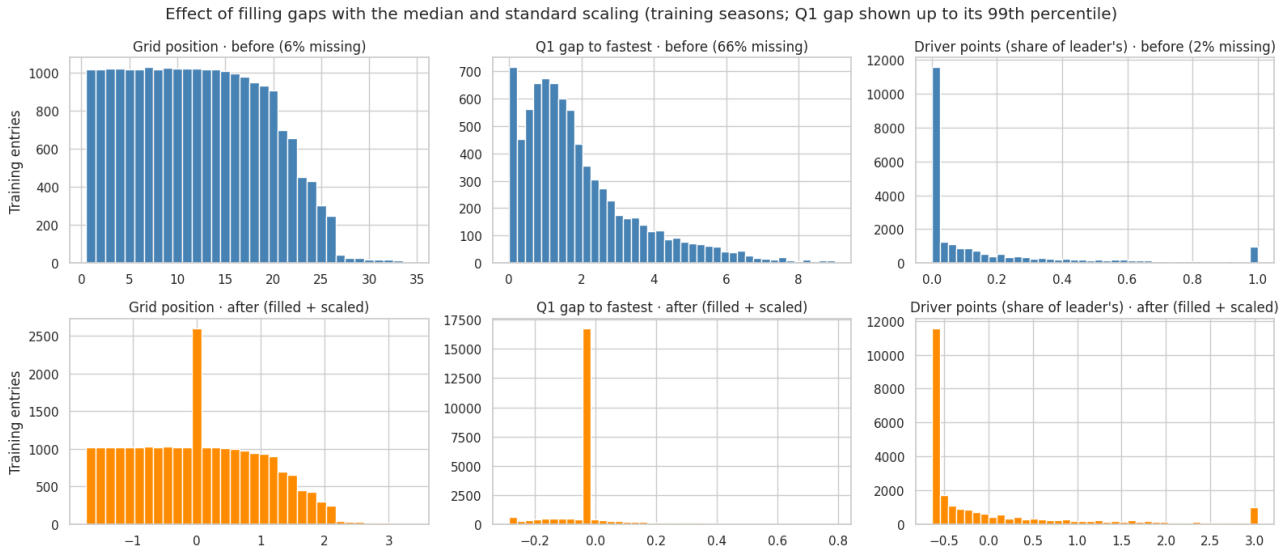


Figure 10. *Three features before (top) and after (bottom) median imputation and standard scaling, on the training seasons. The tall bars are imputed values.*

7.3 Preprocessing-order experiment and choice of training window

Logistic regression and gradient boosting were trained for every combination of 9 training start years (1950, 1960, 1970, 1980, 1990, 2000, 2003, 2010, 2014), 3 feature sets and 2 preprocessing orders — **A:** impute then scale, **B:** scale then impute — 108 models, each scored on validation. 2003 was included because it is the first season with complete qualifying data.

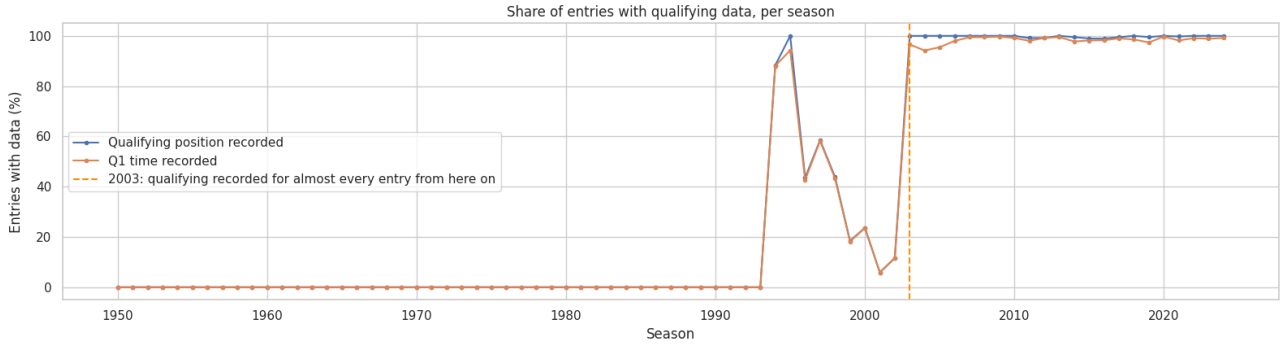


Figure 11. *Share of entries with qualifying data per season: none before 1994, patchy in 1996–2002, complete from 2003.*

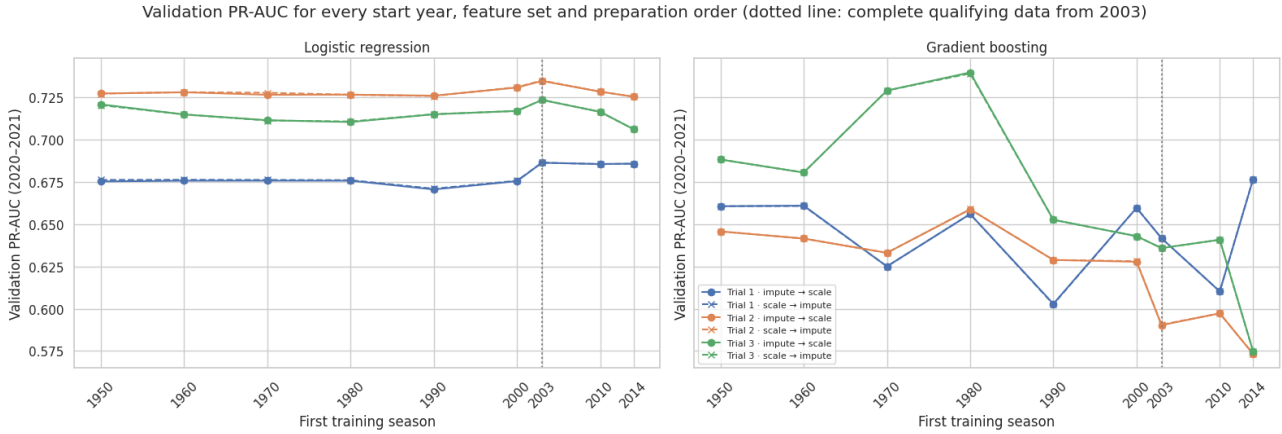


Figure 12. *Validation PR-AUC for every start year, feature set (colour) and preprocessing order (solid = A, dashed = B).*

Order result: across 27 paired runs per model, switching the order changed validation PR-AUC by at most **0.001** for both models (average difference 0). This is expected: median imputation and

scaling only shift and stretch each feature, so both orders give the models the same information. Order A was used everywhere afterwards.

Start year: validation holds only 44 races, so the races were re-drawn at random 500 times (race-level bootstrap) and every start year was scored on the same re-draws. 1980 had the best average validation PR-AUC (0.695, averaged over both models and all trials); only 1960 was statistically as good, so the later year, 1980, was chosen. Logistic regression on its own preferred 2003 and gradient boosting 1980; the combined score decided. At 1980, Trial 3 scored best (0.725 vs 0.693 for Trial 2 and 0.666 for Trial 1). The trade-off is that about half of the 1980–2019 training rows have no qualifying data.

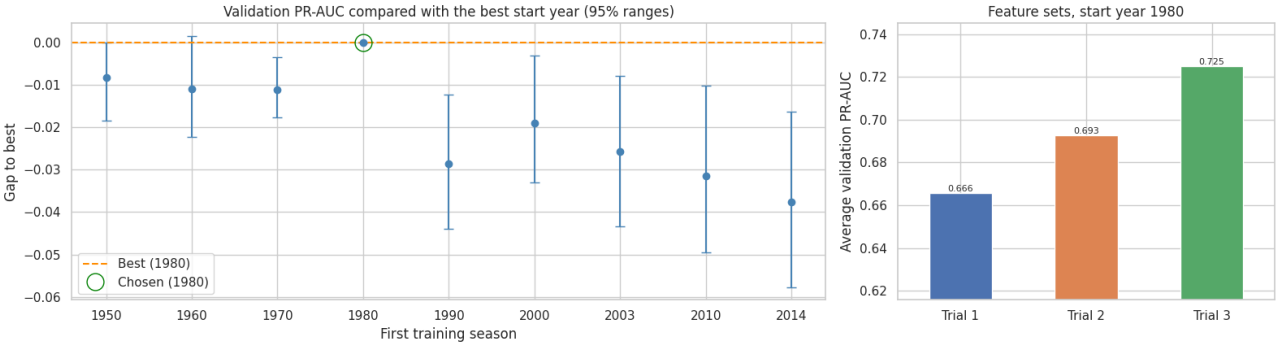


Figure 13. *Left: gap to the best start year with 95% ranges from the race-level re-draws. Right: feature sets at the chosen start year.*

8. Predictive models and results

8.1 Models

- **Baseline:** logistic regression using only the grid position — the simplest reasonable model. Every other model must beat it to justify its complexity.
- **Logistic regression** (all 10 features): weights each feature and turns the sum into a probability. Stable and interpretable, but cannot learn interactions on its own.
- **Gradient boosting:** many small decision trees, each correcting the previous ones. Four settings were compared on validation (learning rate 0.05/0.1 × depth 2/3, 200 trees); the best — learning rate 0.05, depth 2 (validation PR-AUC 0.724) — was used for the final model. The deepest, fastest version fitted training best but generalised worse (train 0.752, validation 0.712).
- **Shallow neural network (Keras):** 10 inputs → 32 ReLU → 20% dropout → 16 ReLU → 1 sigmoid; Adam, binary cross-entropy, batches of 64, up to 100 epochs. Two versions were trained; early stopping (patience 10 on validation loss) stopped at epoch 63 with validation PR-AUC 0.756 (vs 0.752 without) and was kept.

All final models use the same rows (1980–2019), the same 10 features and the same preprocessing, so their scores are directly comparable.

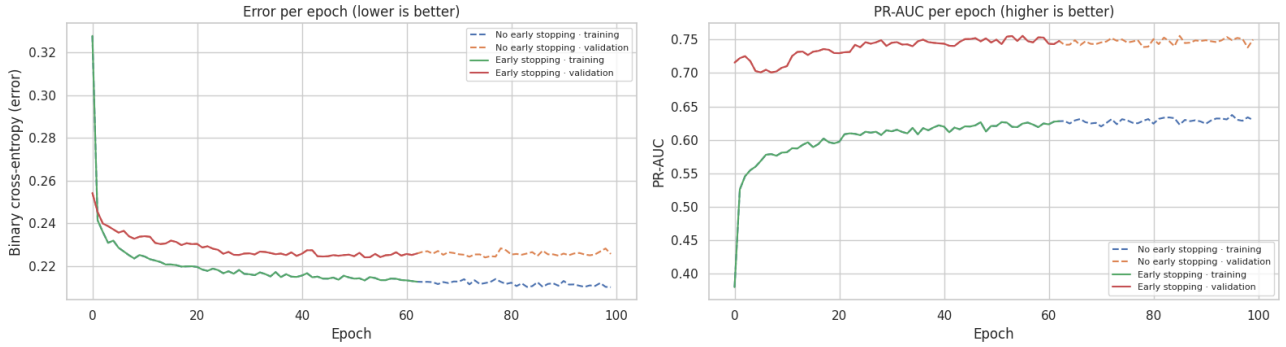


Figure 14. *Neural network training curves: loss and PR-AUC on training and validation for both versions. Validation loss flattens while training loss keeps falling slightly; early stopping ends training there.*

8.2 Metrics

Primary metric: PR-AUC — it measures how well the model ranks real podium finishers at the top, and is more informative than ROC-AUC when positives are rare (about 1 in 7). **Secondary metric: F1** — it scores the final yes/no answer at a probability threshold. ROC-AUC, precision and recall are also reported. Each model's threshold is the value with the best F1 on validation and is reused unchanged on test. No setting was changed after the test results were seen.

Model	Set	PR-AUC	ROC-AUC	F1	Precision	Recall	Threshold
Baseline (grid only)	Train	0.538	0.902	0.581	0.581	0.581	0.36
	Validation	0.639	0.909	0.692	0.692	0.692	
	Test	0.647	0.913	0.649	0.650	0.647	
Logistic regression	Train	0.605	0.916	0.594	0.596	0.591	0.40
	Validation	0.710	0.923	0.709	0.709	0.709	
	Test	0.676	0.930	0.655	0.607	0.711	
Gradient boosting	Train	0.655	0.927	0.622	0.512	0.790	0.23
	Validation	0.724	0.923	0.699	0.613	0.812	
	Test	**0.715**	**0.937**	**0.672**	0.561	0.838	
Shallow neural network	Train	0.641	0.928	0.620	0.589	0.656	0.37
	Validation	**0.756**	0.930	**0.710**	0.719	0.701	
	Test	0.698	0.935	0.662	0.652	0.672	

Table 6. *Model comparison on the training, validation and unseen test seasons (all five metrics).*

8.3 Interpretation

- **All three models beat the grid-only baseline** on the test seasons (PR-AUC 0.676–0.715 vs 0.647), so the additional features and model complexity are justified.
- **The neural network was best on validation** (PR-AUC 0.756, F1 0.710); **gradient boosting was best on test** (PR-AUC 0.715, F1 0.672). With about 40–70 races per set, differences of a few hundredths are within chance, so the three models perform similarly.
- **The biggest validation-to-test drop** is for the neural network (0.059). The test seasons (2022+) are a new technical era, so some drop is expected.
- **ROC-AUC is 0.91–0.94 for every model, including the baseline**, because ranking back-markers below front-runners is easy; this is why PR-AUC is the primary metric.
- **Training scores are lower than validation scores** because the training seasons include the 1980s–2000s, with larger fields and less predictable podiums — not because of a bug.

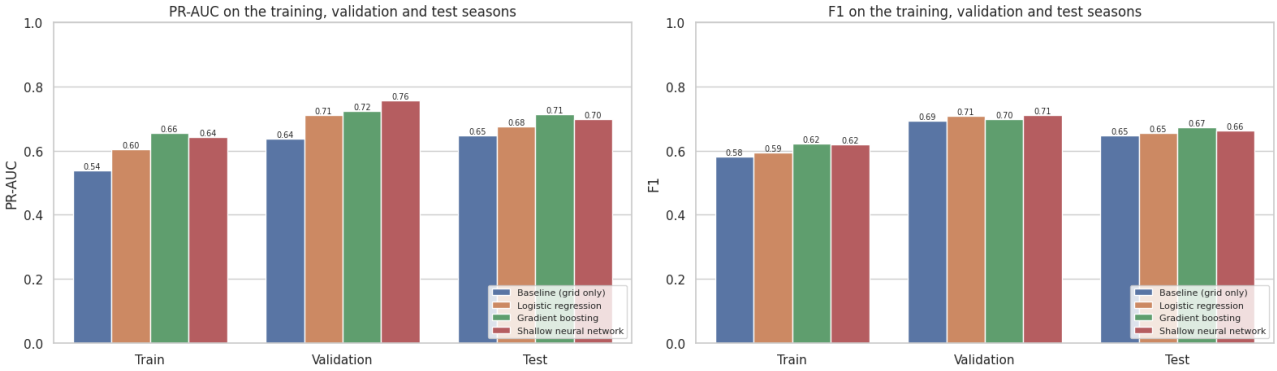


Figure 15. *PR-AUC and F1 for each model on the training, validation and test seasons.*

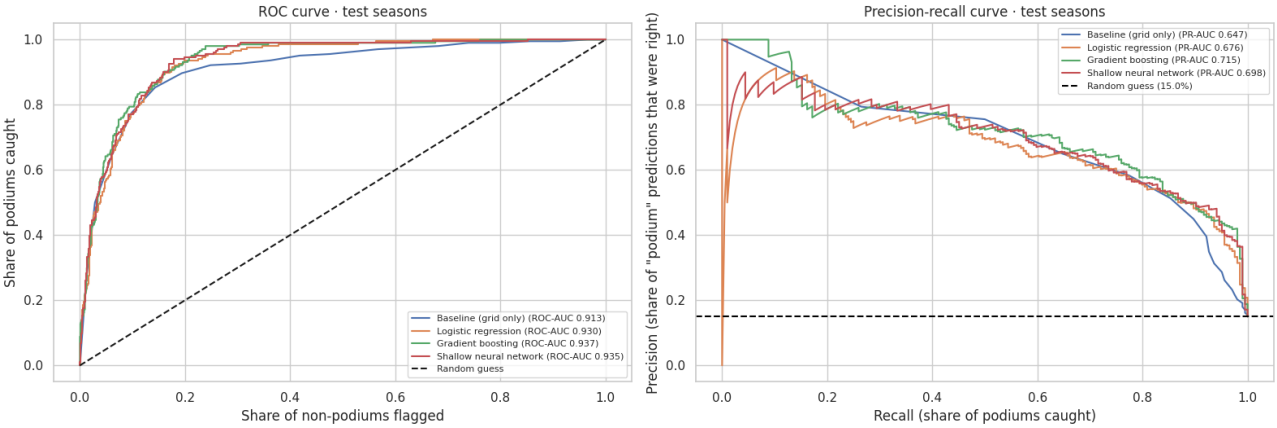


Figure 16. *ROC and precision-recall curves on the test seasons.*

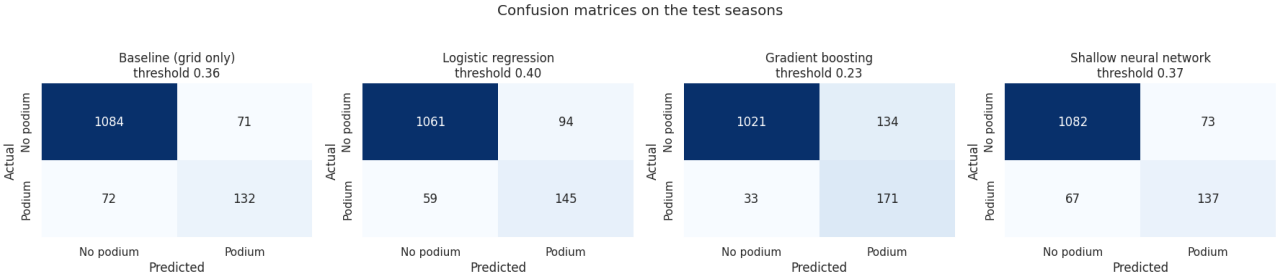


Figure 17. *Confusion matrices on the test seasons at each model's validation threshold.*

9. Feature-group ablation

Each feature group was removed in turn, and all three final models were retrained with the same rows, preprocessing and settings. The validation drop decides; the test drop is a check only. Section 6 justified the features before training; the ablation measures their actual contribution after training.

Removed group	Logistic regression	Gradient boosting	Neural network	Reading
Driver form	+0.043	+0.058	+0.060	All models clearly rely on it
Grid	+0.050	+0.017	+0.036	All models clearly rely on it
Circuit	−0.016	+0.024	+0.016	Models disagree: the non-linear models use it
Qualifying	+0.000	+0.011	+0.010	Small; mostly duplicated by grid
Team form	−0.013	−0.014	+0.008	No gain; overlaps with driver form

Table 7. *Drop in validation PR-AUC when each group is removed (positive = the model got worse).*

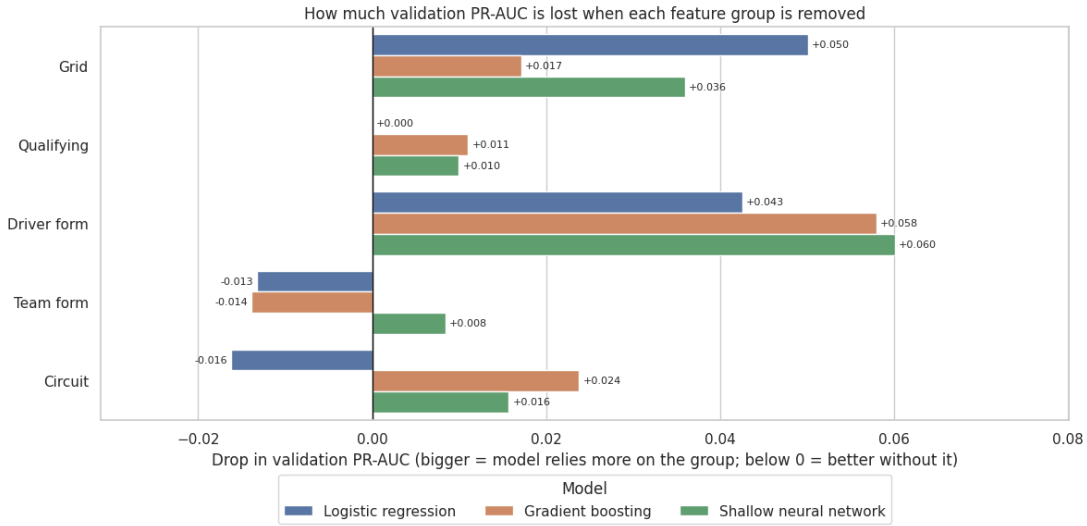


Figure 18. *Feature-group ablation for all three models (validation PR-AUC drop).*

Grid and driver form carry most of the signal. Qualifying adds little once the grid is known (grid and qualifying correlate at 0.90), and team form adds nothing beyond driver form (their points shares correlate at 0.94). The circuit group helps the tree and network models but not logistic regression; on test, removing it hurt gradient boosting strongly (0.715 → 0.439), which suggests that model uses circuit information in combination with other features.

10. Model explainability (XAI)

Explanations are not causes. Permutation importance, SHAP and LIME show which features the models *use* to make predictions. A feature being important to a model does **not** prove that it causes podium finishes. For example, a high points share does not make a car faster; it is a sign of a strong car and driver, which the model uses as a clue.

All explanations use the final trained models without retraining, on the test seasons. The detailed SHAP and LIME analysis uses the model with the best validation PR-AUC, the shallow neural network.

10.1 Global: permutation importance

Each feature is shuffled (10 repeats) so that it no longer matches its driver, and the drop in test PR-AUC is measured. Grid position dominates for every model (0.29–0.33 PR-AUC lost). Next come team past podium rate and team points share (logistic regression, neural network) and driver past podium rate and qualifying position (gradient boosting). Features near 0 can be shuffled without harm, either because the model ignores them or because a correlated feature carries the same information.

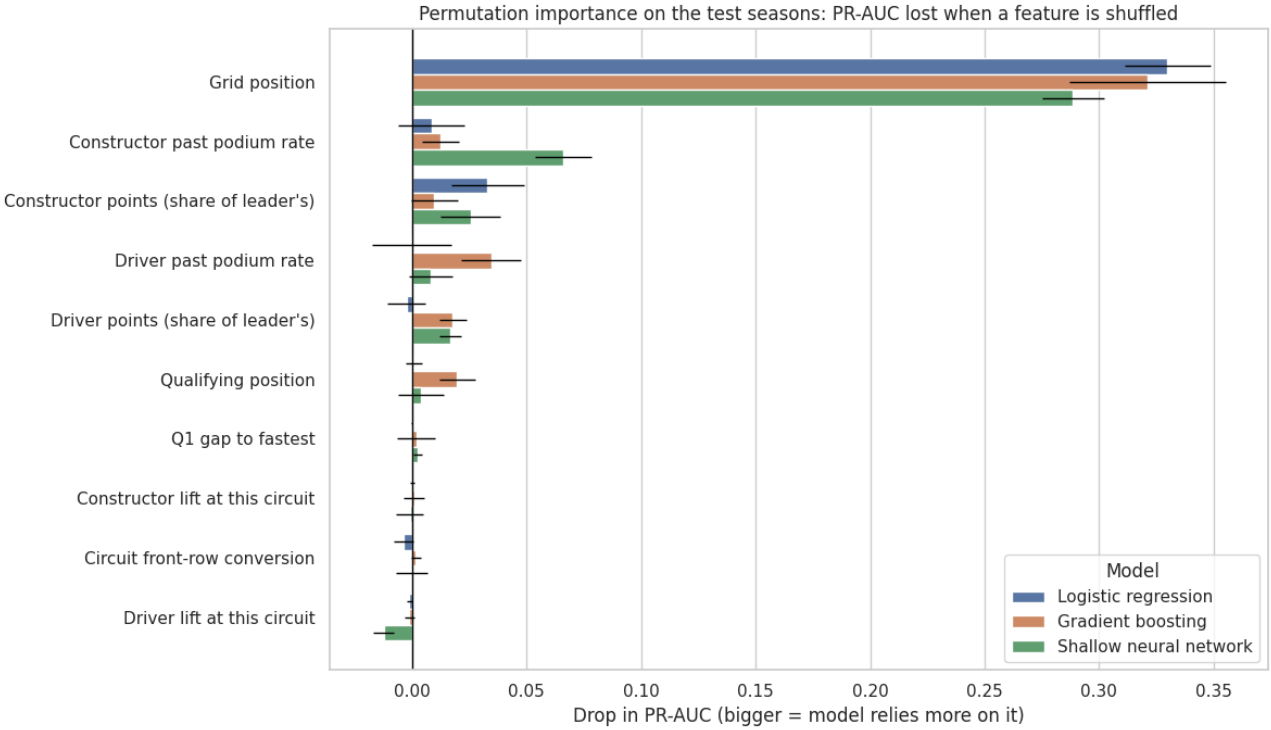


Figure 19. *Permutation importance on the test seasons for all three models.*

10.2 Global: SHAP

SHAP splits each prediction into contributions from each feature, starting from the average prediction (0.142 for the neural network); the contributions add up exactly to the model's output. Over 300 test entries, the largest average contributions for the neural network are grid position (±0.088), team past podium rate (±0.037), team points share (±0.024) and driver past podium rate (±0.021). The directions are intuitive: a lower grid number and higher form values raise the probability. SHAP and permutation importance agree that grid position is the top feature.

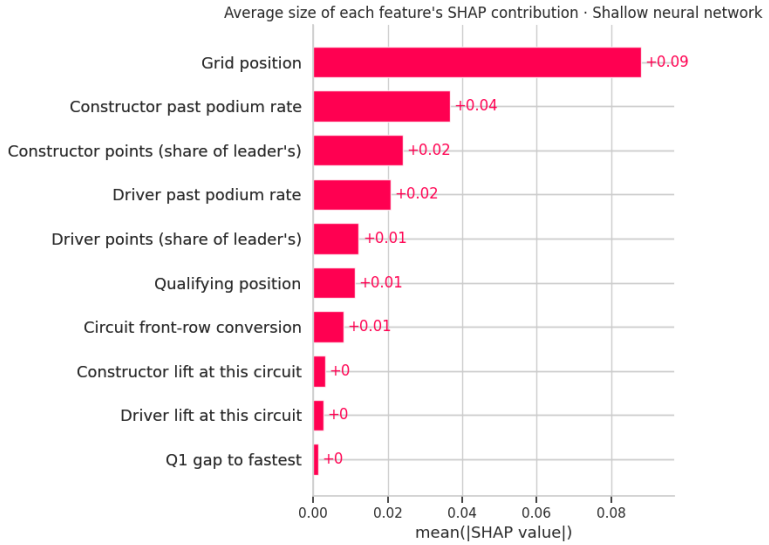


Figure 20. *Average size of each feature's SHAP contribution (neural network).*

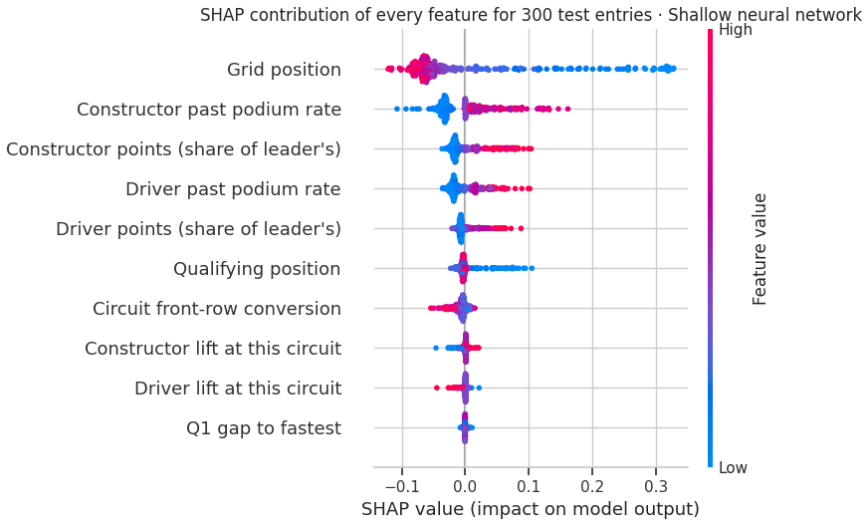


Figure 21. *SHAP beeswarm: one dot per test entry per feature; right = raised the probability, red = high feature value.*

10.3 Local: SHAP waterfall and LIME

Two test entries were chosen automatically: the podium finisher the model was most confident about, and the podium finisher it rated lowest.

- **Clear call — Max Verstappen, 2024 Bahrain GP** (grid 1, predicted 0.902). SHAP: grid position +0.359 and team past podium rate +0.196 push the probability up. LIME agrees: "grid position ≤ 7" (+0.163) and "team past podium rate > 0.25" (+0.062).
- **Surprise — Pierre Gasly, 2024 São Paulo GP** (grid 13, predicted 0.001, but finished on the podium). SHAP: grid position (−0.058) and a weak team record (−0.032) push it down; LIME: "12 < grid ≤ 17" (−0.063). The model only knows the pre-race situation; podiums that come from weather, safety cars or retirements ahead cannot be foreseen.

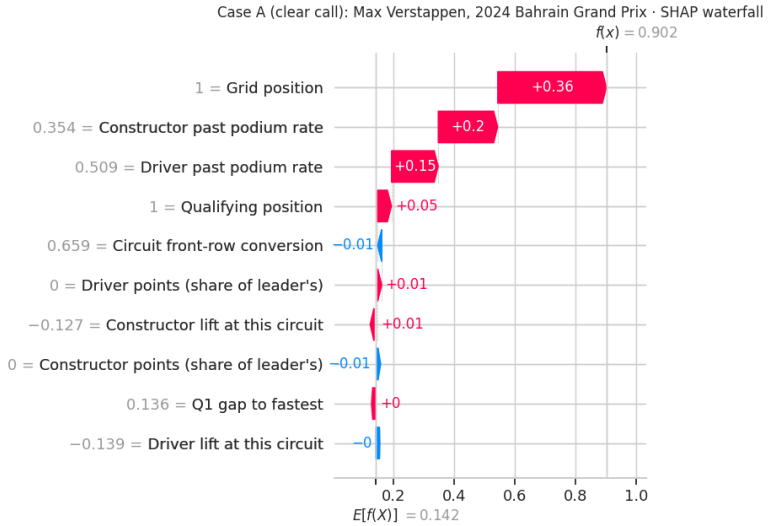


Figure 22. *SHAP waterfall for the clear call: from the average prediction to 0.902.*

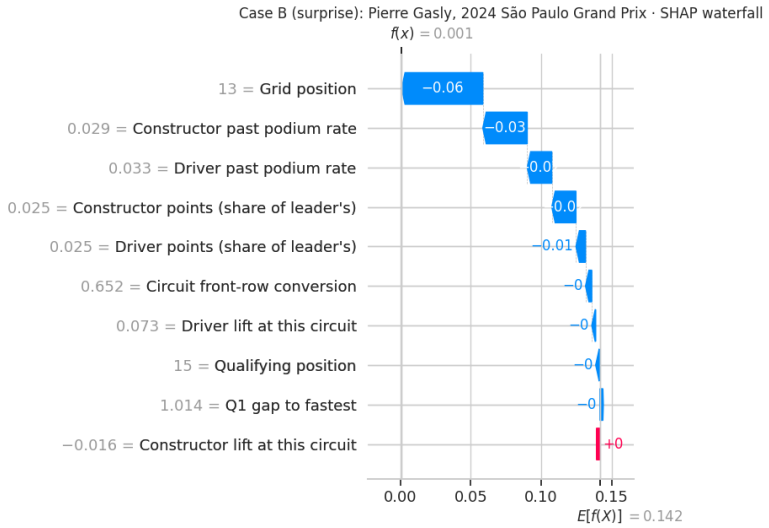


Figure 23. *SHAP waterfall for the surprise podium.*

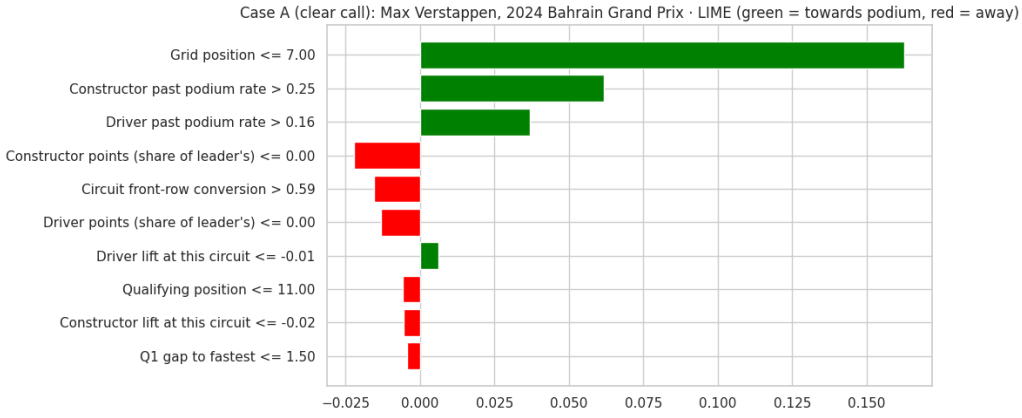


Figure 24. *LIME explanation for the clear call (green = towards podium).*

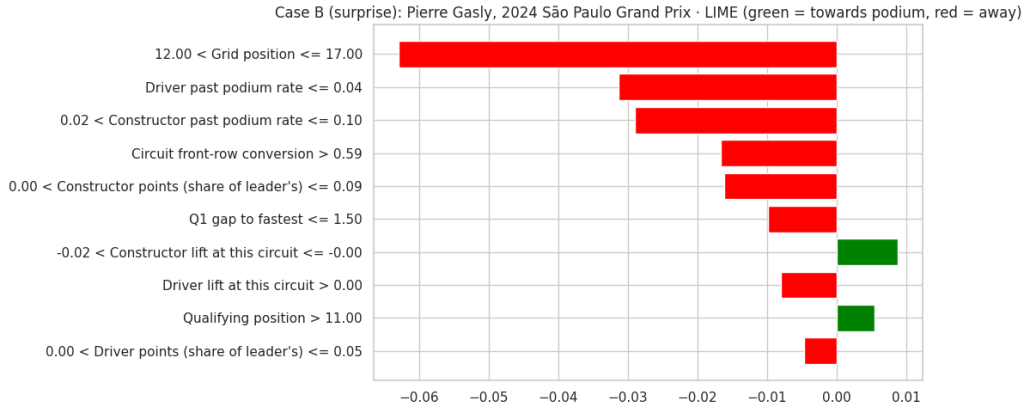


Figure 25. *LIME explanation for the surprise podium.*

SHAP gives exact additive contributions; LIME fits a simple model around one entry. Their numbers differ, but both point to the same main features, which makes the explanations more trustworthy.

11. Inference on new records

The inference function *predict_podium(record)* takes a dictionary with any of the 10 features, rejects unknown feature names, fills missing features with the training median (through the model's own preprocessing), treats grid 0 as missing, and returns the podium probability, the yes/no prediction at the model's validation threshold and the list of features that had to be filled in. It uses the final model with the best validation PR-AUC (the shallow neural network, threshold 0.37).

Record	Missing features	Podium probability	Prediction
Complete (Charles Leclerc, 2022 Bahrain GP, grid 1)	none	0.715	Podium
Partly missing (same entry, no qualifying data)	Qualifying position, Q1 gap	0.430	Podium

Table 8. *Inference demonstrated on a complete and a partly missing record. Leclerc did finish on the podium.*

With qualifying removed, the function still works; the probability falls because the model no longer knows the driver qualified on pole and relies on the median instead.

12. Known limitations

- **Historical qualifying-coverage drift:** no qualifying data before 1994 and patchy data in 1996–2002. With training from 1980, about half of the training rows have median-imputed qualifying values, which hides real differences between drivers in those years.
- **Era sensitivity:** rules, car performance, field sizes and points systems change between eras. The test seasons (2022+) are a new technical era; the points share and past-rate features reduce but do not remove this drift.
- **Small validation and test sets:** 44 and about 68 races. Single scores move by roughly ±0.05 by chance, so close model rankings are not definitive.
- **Model stability:** gradient boosting's scores vary strongly between neighbouring start years and settings; its choice of setting depends on near-ties on validation.
- **Simple preprocessing:** plain median imputation and scaling are sensitive to the extreme Q1 gaps (up to 923%), which compress the useful range of that feature.
- **Pre-race information only:** crashes, mechanical failures, weather, safety cars and strategy decide many podiums and cannot be predicted from these features.

- **Independent predictions:** each driver is scored on their own, so the probabilities in one race need not sum to three podium places.
- **Historical duplicates and definitions:** 1950s shared drives are resolved by result, not by starting car; the home-race mapping excludes dual nationalities; the mechanical classification excludes vague statuses and may miss some failures.
- **Not causal:** neither the feature importance nor the data-engineering associations establish cause and effect.

13. Conclusion

Using only information available before the race, the models identify podium finishers far better than chance and better than a grid-only baseline: test PR-AUC 0.676–0.715 against 0.647. The starting grid is the dominant signal, and driver form adds a clear, independent contribution; qualifying, team form and circuit history add little once those are known. All choices — training window, feature set, preprocessing order, model settings and thresholds — were made on the validation seasons, and the test seasons were used once. The main risks are the shift to a new technical era and the small number of recent races available for evaluation.